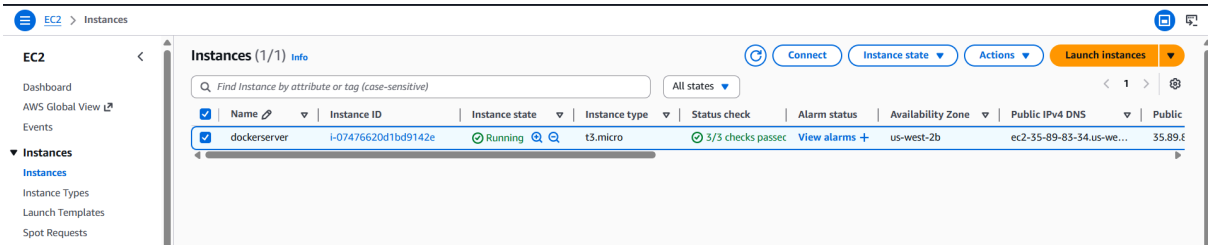
EC2 iNSTANCE CREATED



Docker installed

```
[ec2-user@ip-172-31-16-195 ~]$ sudo su -
[root@ip-172-31-16-195 ~]# sudo yum update -y
sudo yum install docker -y
sudo systemctl start docker
sudo systemctl enable docker
Amazon Linux 2023 Kernel Livepatch repository                                                    261 kB/s |  30 kB     00:00
Dependencies resolved.
Nothing to do.
Complete!
Last metadata expiration check: 0:00:02 ago on Tue Mar 10 04:17:29 2026.
Dependencies resolved.
================================================================================================================================
 Package                                  Architecture                Version                                  Repository                      Size
================================================================================================================================
Installing:
 docker                                   x86_64                      25.0.14-1.amzn2023.0.2                   amazonlinux                     46 M
Installing dependencies:
 container-selinux                        noarch                      4:2.245.0-1.amzn2023                     amazonlinux                     58 k
 containerd                               x86_64                      2.2.1-1.amzn2023.0.1                     amazonlinux                     24 M
 iptables-libs                            x86_64                      1.8.8-3.amzn2023.0.2                     amazonlinux                    401 k
 iptables-nft                             x86_64                      1.8.8-3.amzn2023.0.2                     amazonlinux                    183 k
 libcgroup                                x86_64                      3.0-1.amzn2023.0.1                       amazonlinux                     75 k
 libnetfilter_conntrack                   x86_64                      1.0.8-2.amzn2023.0.2                     amazonlinux                     58 k
 libnfnetlink                             x86_64                      1.0.1-19.amzn2023.0.2                    amazonlinux                     30 k
 libnftnl                                 x86_64                      1.2.2-2.amzn2023.0.2                     amazonlinux                     84 k
 pigz                                     x86_64                      2.5-1.amzn2023.0.3                       amazonlinux                     83 k
 runc                                     x86_64                      1.3.4-1.amzn2023.0.2                     amazonlinux                    3.9 M
```

```
Installed:
  container-selinux-4:2.245.0-1.amzn2023.noarch          containerd-2.2.1-1.amzn2023.0.1.x86_64          docker-25.0.14-1.amzn2023.0.2.x86_64
  iptables-libs-1.8.8-3.amzn2023.0.2.x86_64              iptables-nft-1.8.8-3.amzn2023.0.2.x86_64        libcgroup-3.0-1.amzn2023.0.1.x86_64
  libnetfilter_conntrack-1.0.8-2.amzn2023.0.2.x86_64     libnfnetlink-1.0.1-19.amzn2023.0.2.x86_64       libnftnl-1.2.2-2.amzn2023.0.2.x86_64
  pigz-2.5-1.amzn2023.0.3.x86_64                         runc-1.3.4-1.amzn2023.0.2.x86_64

Complete!
Created symlink /etc/systemd/system/multi-user.target.wants/docker.service → /usr/lib/systemd/system/docker.service.
[root@ip-172-31-16-195 ~]# sudo usermod -aG docker ec2-user
[root@ip-172-31-16-195 ~]# docker --version
Docker version 25.0.14, build 0bab007
[root@ip-172-31-16-195 ~]#
```

Installed Docker Compose

```
[root@ip-172-31-16-195 ~]# sudo curl -L "https://github.com/docker/compose/releases/latest/download/docker-compose-$(uname -s)-$(uname -m)" -o /usr/local/bin/docker-co
mpose
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
  0     0    0     0    0     0      0      0  --:--:-- --:--:-- --:--:--     0
  0     0    0     0    0     0      0      0  --:--:-- --:--:-- --:--:--     0
100 30804k 100 30804k    0     0 50060k      0  --:--:-- --:--:-- --:--:-- 95667k
[root@ip-172-31-16-195 ~]#
```

```
[root@ip-172-31-16-195 ~]# docker-compose --version
Docker Compose version v5.1.0
[root@ip-172-31-16-195 ~]#
```

```
[root@ip-172-31-16-195 ~]# sudo chmod +x /usr/local/bin/docker-compose
[root@ip-172-31-16-195 ~]# docker-compose --version
Docker Compose version v5.1.0
[root@ip-172-31-16-195 ~]# mkdir nginx-custom
cd nginx-custom
[root@ip-172-31-16-195 nginx-custom]# pwd
/root/nginx-custom
[root@ip-172-31-16-195 nginx-custom]# nano nginx.conf
[root@ip-172-31-16-195 nginx-custom]# cat ngnix.conf
cat: ngnix.conf: No such file or directory
[root@ip-172-31-16-195 nginx-custom]# cat nginx.conf
server {
    listen 80;

    location / {
        root /var/opt/nginx;
        index index.html;
    }
}
[root@ip-172-31-16-195 nginx-custom]#
```

Docker File Created

```
[root@ip-172-31-16-195 nginx-custom]# cat nginx.conf
server {
    listen 80;

    location / {
        root /var/opt/nginx;
        index index.html;
    }
}
[root@ip-172-31-16-195 nginx-custom]# nano Dockerfile
[root@ip-172-31-16-195 nginx-custom]# cat Dockerfile
FROM nginx:latest
COPY nginx.conf /etc/nginx/conf.d/default.conf
[root@ip-172-31-16-195 nginx-custom]#
```

Compose file created

```
}
[root@ip-172-31-16-195 nginx-custom]# nano Dockerfile
[root@ip-172-31-16-195 nginx-custom]# cat Dockerfile
FROM nginx:latest
COPY nginx.conf /etc/nginx/conf.d/default.conf
[root@ip-172-31-16-195 nginx-custom]# nano docker-compose.yml
[root@ip-172-31-16-195 nginx-custom]# cat docker-compose.yml
version: '3.8'

services:
  nginx:
    image: wikkics/custom-nginx:v1
    container_name: custom-nginx
    ports:
      - "80:80"
    volumes:
      - /var/opt/nginx:/var/opt/nginx
[root@ip-172-31-16-195 nginx-custom]#
```

Created volume directory

```
[root@ip-172-31-16-195 nginx-custom]# mkdir -p /var/opt/nginx
[root@ip-172-31-16-195 nginx-custom]# nano /var/opt/nginx/index.html
[root@ip-172-31-16-195 nginx-custom]# cat index.html
cat: index.html: No such file or directory
[root@ip-172-31-16-195 nginx-custom]# cat /var/opt/nginx/index.html
<h1>Welcome to Custom Nginx via Docker Compose</h1>
<p>Volume mounted at /var/opt/nginx</p>
[root@ip-172-31-16-195 nginx-custom]#
```

Builded the custom Docker image

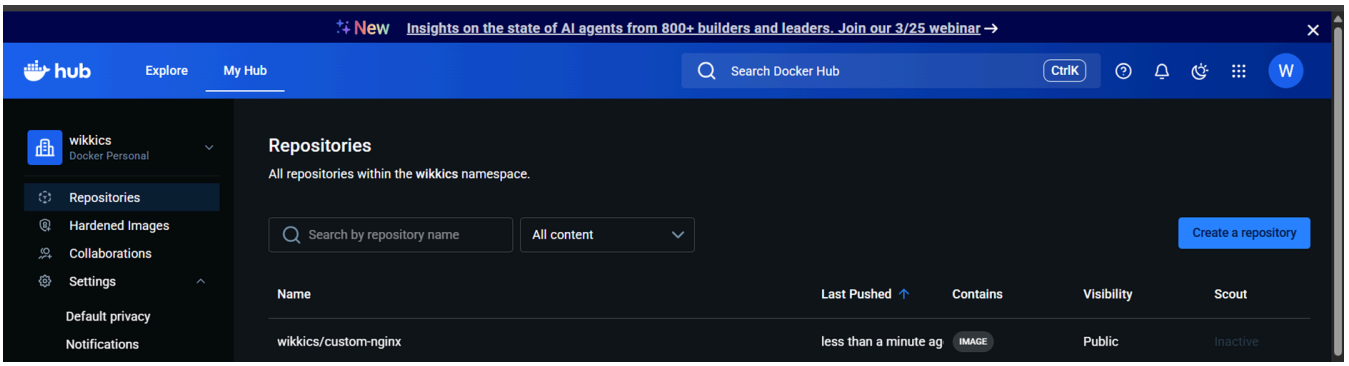
```
[root@ip-172-31-16-195 nginx-custom]# docker build -t wikkics/custom-nginx:v1 .
[+] Building 5.1s (7/7) FINISHED
 => [internal] load build definition from Dockerfile
 => => transferring dockerfile: 162B
 => [internal] load metadata for docker.io/library/nginx:latest
 => [internal] load .dockerignore
 => => transferring context: 2B
 => [internal] load build context
 => => transferring context: 202B
 => [1/2] FROM docker.io/library/nginx:latest@sha256:0236ee02dcbce00b9bd83e0f5fbc51069e7e1161bd59d99885b3ae1734f3392e
 => => resolve docker.io/library/nginx:latest@sha256:0236ee02dcbce00b9bd83e0f5fbc51069e7e1161bd59d99885b3ae1734f3392e
 => => sha256:0236ee02dcbce00b9bd83e0f5fbc51069e7e1161bd59d99885b3ae1734f3392e 10.23kB / 10.23kB
 => => sha256:b7f90eb243ff5e2c4a7dc61768160e888ac459e2c05aeb1b3bdf90920aa9d7e9 2.29kB / 2.29kB
 => => sha256:fd204fe2f75024354b1f979d38cc43def9e049cc2df1cda45074d1b84c4f9b3e 9.09kB / 9.09kB
 => => sha256:206356c42440674ecbdf1070cf70ce8ef7885ac2e5c56f1ecf800b758f6b0419 29.78MB / 29.78MB
 => => sha256:b47f187216b68c3f90b7fb9a947a76d4bd2981a8f469cb6092e7da5d9efea014 33.14MB / 33.14MB
 => => sha256:1ad233904a1166fd3832d9657b163937f66c906795a517a87b516bc3f2481695 626B / 626B
 => => sha256:eedda9fd87862df4b2eccd3da9f2e38122124dec2e4217b530bd48d809b3840b 954B / 954B
 => => sha256:35ff83c394d65bf2c49f844ba18fe2e8121fe07e1f3f92325c05246319b98f7e 405B / 405B
 => => extracting sha256:206356c42440674ecbdf1070cf70ce8ef7885ac2e5c56f1ecf800b758f6b0419
 => => sha256:17d0911eaf62f5c4c34e54943d30dbb0d86b8ad614c80ef5ee0249ff54a3373e 1.21kB / 1.21kB
 => => sha256:df0b66c867e491634d63f065ef5ea556cba20a97761039f9e9d5af71442f8fbf 1.40kB / 1.40kB
 => => extracting sha256:b47f187216b68c3f90b7fb9a947a76d4bd2981a8f469cb6092e7da5d9efea014
 => => extracting sha256:1ad233904a1166fd3832d9657b163937f66c906795a517a87b516bc3f2481695
 => => extracting sha256:eedda9fd87862df4b2eccd3da9f2e38122124dec2e4217b530bd48d809b3840b
 => => extracting sha256:35ff83c394d65bf2c49f844ba18fe2e8121fe07e1f3f92325c05246319b98f7e
 => => extracting sha256:17d0911eaf62f5c4c34e54943d30dbb0d86b8ad614c80ef5ee0249ff54a3373e
 => => extracting sha256:df0b66c867e491634d63f065ef5ea556cba20a97761039f9e9d5af71442f8fbf
 => [2/2] COPY nginx.conf /etc/nginx/conf.d/default.conf
 => exporting to image
 => => exporting layers
 => => writing image sha256:26bec24669772279ab1b4934247763a5be9f711500e45e48a0123db73f1b5b73
 => => naming to docker.io/wikkics/custom-nginx:v1
[root@ip-172-31-16-195 nginx-custom]# docker images
REPOSITORY             TAG       IMAGE ID       CREATED          SIZE
wikkics/custom-nginx   v1        26bec2466977   29 seconds ago   161MB
[root@ip-172-31-16-195 nginx-custom]#
```

Running container using Docker Compose

```
[root@ip-172-31-16-195 nginx-custom]# docker images
REPOSITORY          TAG       IMAGE ID       CREATED          SIZE
wikkics/custom-nginx   v1       26bec2466977   29 seconds ago   161MB
[root@ip-172-31-16-195 nginx-custom]# docker-compose up -d
WARN[0000] /root/nginx-custom/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion
[+] up 2/2
 ✓ Network nginx-custom_default Created                                                                    0.2s
 ✓ Container custom-nginx       Started                                                                    0.6s
[root@ip-172-31-16-195 nginx-custom]# docker ps
CONTAINER ID   IMAGE                  COMMAND                  CREATED          STATUS          PORTS                                   NAMES
f1898e9500ac   wikkics/custom-nginx:v1   "/docker-entrypoint.…"   15 seconds ago   Up 15 seconds   0.0.0.0:80->80/tcp, :::80->80/tcp   custom-nginx
[root@ip-172-31-16-195 nginx-custom]#
```

Testing Nginx

```
[root@ip-172-31-16-195 nginx-custom]# curl http://localhost
<h1>Welcome to Custom Nginx via Docker Compose</h1>
<p>Volume mounted at /var/opt/nginx</p>
[root@ip-172-31-16-195 nginx-custom]#
```

Pushing image to Docker Hub

```
[root@ip-172-31-16-195 nginx-custom]# docker login
Log in with your Docker ID or email address to push and pull images from Docker Hub. If you don't have a Docker ID, head over to https://hub.docker.com/ to create one.
You can log in with your password or a Personal Access Token (PAT). Using a limited-scope PAT grants better security and is required for organizations using SSO. Learn more at https://do
cs.docker.com/go/access-tokens/

Username: wikkics
Password:
WARNING! Your password will be stored unencrypted in /root/.docker/config.json.
Configure a credential helper to remove this warning. See
https://docs.docker.com/engine/reference/commandline/login/#credentials-store

Login Succeeded
[root@ip-172-31-16-195 nginx-custom]# docker push wikkics/custom-nginx:v1
The push refers to repository [docker.io/wikkics/custom-nginx]
2bddf71c2c54: Pushed
87026f913c5d: Mounted from library/nginx
d120c727e5e3: Mounted from library/nginx
a821e7df95cb: Mounted from library/nginx
b9ae255e9e51: Mounted from library/nginx
a507d5bc786a: Mounted from library/nginx
7351f826b314: Mounted from library/nginx
a257f20c716c: Mounted from library/nginx
v1: digest: sha256:841c18be6043ce5bc5b80946f9bcffee8ed49164338ed30e861a30211cfbd89a size: 1985
[root@ip-172-31-16-195 nginx-custom]#
```